ENGR 225: Mechanics 1

Homework Coversheet

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DATE: 6 November 2026

ASSIGNMENT: Design Project, Part 1

Level of involvement of AI, LLMs, or related tools in the completion of this assignment:

☐ None

☐ Minor

☐ Major

☐ Extensive

Mark the box that applies (e.g., replace ☐ with ■, or type an X beside it, before compiling/printing). Your disclosure does not affect your grade – it keeps you accountable for the work you are doing.

ENGR 225 Design Project, Part 1: Roadway Selection

Sean Balbale Assigned: 26 Oct 2026 Due: 6 Nov 2026

1 Design Basis and Classification

The roadway carries two lanes of traffic in each direction across a 60 m canyon span with no pedestrian accommodation, and it is meant to be comparable to Route 2 outside Hartford. I classify it as an urban principal arterial *freeway*: divided, full control of access, and two through lanes per direction. Route 2 over Griswold Street in Glastonbury, a six-lane section of the same corridor, carries an average daily traffic of 74,400 vehicles with 6% trucks (2021) [1]. I use that count as the design volume, which works out to roughly 4,460 trucks per day.

Because the roadway sits on a bridge, two layers of standards govern it. Geometric criteria (lane and shoulder widths) come from AASHTO's freeway policy, and the deck section (slab, waterproofing, wearing surface) comes from the CTDOT Bridge Design Manual and CTDOT pavement guidance. The bridge carries the full approach width rather than narrowing over the span.

2 Geometry and Materials

Cross-Section Widths AASHTO requires a minimum freeway lane width of 12 ft and a minimum shoulder width of 10 ft, except for the left (median-side) shoulder of a 4-lane freeway, which may be 4 ft [2]. I use those minimums directly, since minimizing material is a stated design goal and every foot of width carries through to every quantity below. Barriers are a 32 in concrete parapet on each edge and a 42 in median barrier. Barrier base widths of 21 in (parapet) and 24 in (median) are my assumptions, not CTDOT plate values.

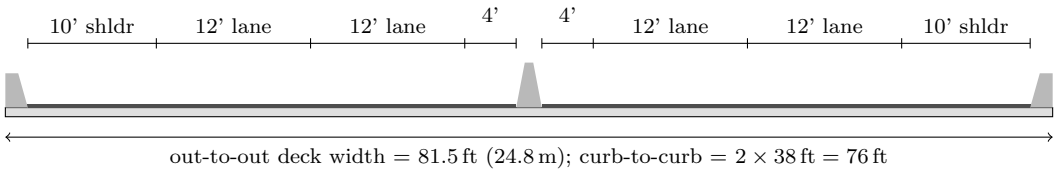


Figure 1: Roadway cross-section looking along the span: 3 in HMA on membrane over an 8.5 in reinforced concrete slab (not to scale vertically).

Deck Section The deck section follows Connecticut practice from the bottom up:

- **Structural slab:** cast-in-place reinforced concrete, 8.5 in (216 mm) thick. This is the CTDOT minimum for a cast-in-place deck, with 2.5 in of cover over the top reinforcing mat [3]. Reinforcement is taken as 150 kg/m^3 of concrete, a typical value for a two-mat deck; the real figure comes out of the Part 2 load analysis.
- **Waterproofing:** CTDOT protects its decks with a waterproofing membrane under a bituminous riding surface, and has shifted mostly to spray-applied cold liquid elastomeric membranes [4].

- **Wearing surface:** 3 in of hot-mix asphalt (HMA), CTDOT's standard overlay thickness on bridge decks: a 1 in HMA S0.25 lift directly over the membrane and a 2 in HMA S0.5 top course [4].

Quantities With $L = 60\,\text{m}$, out-to-out width $W = 24.84\,\text{m}$, and curb-to-curb width $w_c = 23.16\,\text{m}$:

$$\begin{aligned} V_{\text{deck}} &= L\,W\,t_s = (60\,\text{m})(24.84\,\text{m})(0.216\,\text{m}) = 322\,\text{m}^3 \\ V_{\text{barrier}} &= L\,(2A_p + A_m) = (60\,\text{m})(2 \times 0.279\,\text{m}^2 + 0.434\,\text{m}^2) = 59.5\,\text{m}^3 \\ m_{\text{rebar}} &= (381\,\text{m}^3)(150\,\text{kg/m}^3) = 57.2\,\text{t} \\ m_{\text{HMA}} &= L\,w_c\,t_o\,\rho_{\text{HMA}} = (60\,\text{m})(23.16\,\text{m})(0.076{,}2\,\text{m})(2.35\,\text{t/m}^3) = 249\,\text{t} \end{aligned}$$

Total concrete is $381\,\text{m}^3$ ($499\,\text{yd}^3$) and membrane area is $1{,}390\,\text{m}^2$.

3 Environmental Impact

Embodied Carbon (A1–A3, Cradle to Gate) I quantify global warming potential (GWP) using environmental product declarations (EPDs), which report kg CO_2e per unit of material through extraction, transport to the plant, and manufacturing. Ready-mix 4,000 psi mixes from a Connecticut-area producer range from $265\,\text{kg/m}^3$ to $335\,\text{kg/m}^3$ across the mixes in a single NRMCA-verified EPD [5]; I use the $300\,\text{kg/m}^3$ midpoint. The CRSI industry-average EPD gives 854 kg CO_2e per metric ton of fabricated rebar [6], and a representative HMA EPD published by FHWA gives 71.1 kg per metric ton [7].

Table 1: Baseline embodied carbon of the roadway materials (A1–A3)

Material	**Quantity**	**GWP Factor**	**CO_2e (t)**
Concrete (deck + barriers)	$381\,\text{m}^3$	$300\,\text{kg/m}^3$	114
Reinforcing steel	57.2 t	854 kg/t	48.8
HMA overlay	249 t	71.1 kg/t	17.7
Total			**181**

Concrete accounts for 63% of the total, and almost all of that comes from portland cement, which releases CO_2 both from kiln fuel and from calcining limestone. The membrane is left out because I could not find a published EPD for a spray-applied bridge membrane; at a few millimeters thick over $1{,}390\,\text{m}^2$, its mass is small relative to the slab.

Other Impacts Carbon is not the only cost. The concrete and HMA together contain several hundred tonnes of quarried aggregate, which carries land disturbance, dust, and truck traffic at the quarry. The A1–A3 figures above also stop at the plant gate, so delivery to the site is not counted; a distant plant with a lower EPD value can end up worse than a nearby one once hauling is included. Specifying recycled asphalt pavement (RAP) in the HMA reduces both virgin aggregate and binder demand.

Alternatives and Perspective Table 2 compares the baseline against material substitutions that do not change the geometry.

Table 2: Embodied carbon of alternative material choices

Option	Total CO_2e (t)	Change
Baseline (Table 1)	181	–
Slag-cement concrete, 40–50% lower concrete GWP [8]	124–135	−25 to −32%
EAF rebar on renewable power, 300 kg/t [9]	149	−18%
3 in concrete wearing surface instead of HMA	195	+8%
Slag concrete (45%) + low-carbon rebar	98	−46%

Two results stand out. First, HMA is the correct wearing surface on carbon grounds as well as CTDOT practice: replacing it with 106 m^3 of extra concrete adds 14 t. Second, the cement content of the slab is the single biggest lever. A 60% slag mix approved for Australian infrastructure cut embodied carbon by an estimated 40–50% [8]. The trade-off is slower early strength gain, which affects the construction schedule but not the finished deck.

For scale, EPA puts the average passenger vehicle at about 400 g CO_2 per mile [10]. At 74,400 vehicles per day, traffic crossing this 60 m span emits roughly

$$(74{,}400)\left(\frac{60\,\text{m}}{1{,}609\,\text{m/mi}}\right)(0.400\,\text{kg/mi}) = 1.11\,\text{t}\,CO_2/\text{day},$$

so the baseline materials equal about 163 days of tailpipe emissions on the bridge itself. This overstates the ratio slightly, since trucks emit more per mile than cars. The materials are a one-time cost against a service life measured in decades, but they are the part of the footprint the designer directly controls.

4 Raw Material Cost

Table 3: Raw material cost estimate (2026 USD, delivered, excluding labor and formwork)

Material	Quantity	Unit Price	Cost
Ready-mix concrete	499 yd^3	\$185/yd^3 [11]	\$92,300
Reinforcing steel	57.2 t	\$1,100/t [12]	\$62,900
HMA (S0.25 + S0.5)	274 short tons	\$130/ton [13]	\$35,700
Total			**\$191,000**

The concrete price is the top of the \$155–\$185/yd^3 Northeast range [11], since a bridge deck mix is air-entrained and higher strength than residential concrete. The rebar price rounds up from the US ex-works average of \$1,058/t at the start of 2026 [12] to allow for fabrication. The HMA price is a CTDOT bid price, \$129–\$135/ton for HMA S0.5 on a recent bridge contract [13]; it includes placement, so it is an upper bound on material cost. Membrane material is excluded for the same reason as in Section 3. For comparison, the same contract bid Class F deck concrete at \$700–\$915/yd^3 in place [13]. Raw concrete is roughly a quarter of the installed price; forming, placing, finishing, and curing a deck over a canyon make up the rest.

5 Conclusion

The roadway is a 4-lane urban freeway deck, 60 m long and 24.8 m wide out-to-out:

1. **Geometry:** four 12 ft lanes, 10 ft right and 4 ft left shoulders, 76 ft curb-to-curb, per AASHTO freeway minimums. Widening the left shoulders to 10 ft would add 15% to every quantity in this report.
2. **Materials:** 8.5 in cast-in-place reinforced concrete slab, waterproofing membrane, and 3 in HMA overlay, per the CTDOT Bridge Design Manual and pavement guidance.
3. **Environmental impact:** 181 t CO_2e cradle-to-gate, 63% of it from concrete. Slag cement and low-carbon rebar together cut that to about 98 t.
4. **Cost:** about $191,000 in raw materials, with concrete the largest line at 48%.

The slab thickness and rebar density are the two values most likely to change once Part 2 sizes the truss and the deck span between supports.

AI Use I used Claude (Anthropic) to locate the CTDOT, FHWA, and EPD sources listed below, to run the quantity and carbon arithmetic, and to draft this report from those numbers. I checked each design value against its cited source and made the classification, shoulder-width, and material decisions myself.

References

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[11] "Average concrete cost by state in 2026," UseCalcPro. [Online]. Available: https://usecalcpro.com/blog/average-concrete-cost-by-state-2026

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[13] Connecticut Department of Transportation, "Tabulation of bids, State Project 0155-0169." [Online]. Available: https://portal.ct.gov/-/media/dot/documents/dcontracts/01550169bidtabspdf.pdf